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Multi-city bike share Weather associations and daily time structure

STA5001 Bayesian & Computational Statistics

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Three questions

01 Precipitation

How large is the association after calendar and seasonal adjustment?

How much does it vary across cities?

02 Temperature

What are the direction, shape and uncertainty of the association?

How does the shape change across specifications?

03 Time structure

How do calendar, trend and AR(1) affect diagnostics and prediction?

Does the final window support extending the model?

From the midterm to the final project

Same panel; added hierarchical estimation, time structure and final-window evaluation.

Observational associations; predictions condition on realised weather. Scope excludes causal effects, climate adaptation and operational forecasting.

Data and evaluation design

Frozen panel

Source	Mendeley Data, CC BY 4.0
DOI	`10.17632/2nxpvtz935.1`
Full panel	29 cities, 10,578 city-days
Association	25 continuous systems, 9,120
Prediction	23 cities covering all of 2017, 8,393

Response $y = \log(1 + \text{usage})$. UTCI is a thermal-stress index, not air temperature. Rainfall enters as $\log(1 + \text{mm})$ minus the city mean.

Three cutoffs · 56-day blocks

Window	Cutoff	Training rows
dev 1	2017-07-16	4,529
dev 2	2017-09-10	5,817
final	2017-11-05	7,105

23 cities in every test block. All training dates are at or before the common cutoff. Final-window observations appeared in the midterm full-period exploratory analysis.

Hierarchical Student-t model

Response and conditional mean

$$y_{it} = \log(1 + U_{it})$$

$$y_{it} \sim t_\nu(\mu_{it}, \sigma_i)$$

$$\mu_{it} = \alpha_i + \beta_i^{\text{temp}} T_{it} + \beta_i^{\text{sq}} Q_{it} + \beta_i^{\text{rain}} R_{it} + \text{calendar}_{it} + \text{trend}_{it}$$

City coefficients

$$\beta_i = \beta_{\text{pop}} + \tau_\beta \tilde{\beta}_i$$

Estimated group spread; training-only centring.

Three specifications

M0 weekend, annual harmonic, climate moderators

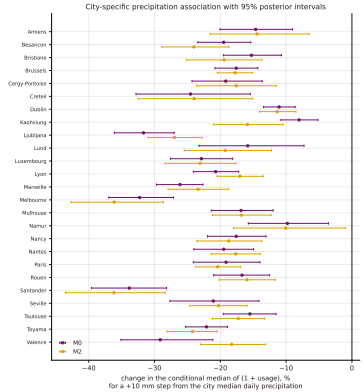
M1 shared weekday effect, shrunk city trend per 100 days

M2 M1 plus city AR(1)

$$e_{it} = \phi_i e_{i,t-1} + \eta_{it}$$

Student-t: heavy tails, including M2 innovations; group spread controls shrinkage. M0/M2 differ in calendar, trend and residual process.

Precipitation: a stable negative association



Negative in every city

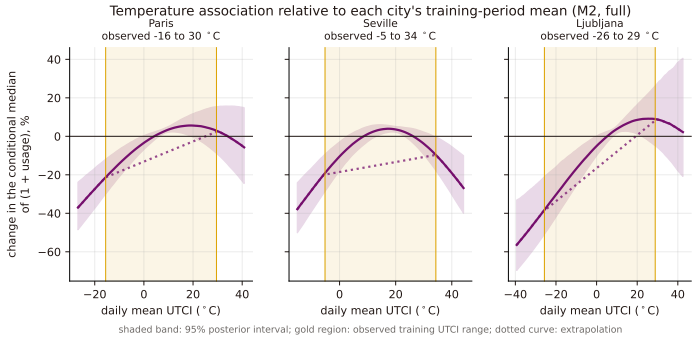
City-equal average:
M2 **-20.21%**, 95% $[-21.37, -18.99]$.
M0 **-19.98%**, 95% $[-21.07, -18.89]$.

Contrast: $+10$ mm from city median daily precipitation. Percent change in the conditional median of $(1 + \text{usage})$.

M2 city medians: -36.18% to -10.05%. Smallest $P(\beta_{\text{rain}} < 0)$: 0.984.

Each specification uses its own saved fit. City ordering has not been tested pairwise.

Temperature: positive association, uncertain peak



Curves: M2 full-period fit. M0/M2 contrasts use the same 25 cities and window. Peak location changes with the specification.

UTCI $+10°$C

M0 **5.22%**
95% $[3.67, 6.76]$.

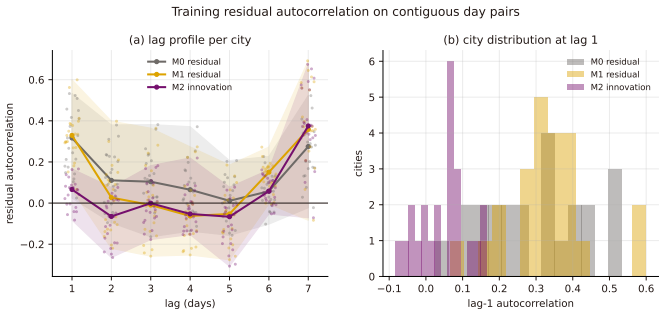
M2 **6.16%**
95% $[4.12, 8.32]$.

Interior peak, $P > 0.95$:
M0 15/25; M2 5/25.

$+10°$C leaves the observed range in 1 city: Kaohsiung.

Separate fits; no single optimal temperature is identified.

Sampling reliability and daily dependence



M0/M1: fitted residuals. M2: innovations.
Lower innovation correlation does not establish the contribution of AR(1) alone; the specifications change several structural components.

All 11 core fits passed

Worst $\widehat{R}$	1.00894
Smallest bulk ESS	426.7
Smallest tail ESS	668.9
Divergences / depth hits	0 / 0

Four chains, 3,000 warmup and 3,000 retained draws; seed 27.

Dev 1: median lag-one

M0 fitted residual	0.317
M1 fitted residual	0.329
M2 innovation	0.066

Sampling, model fit and prediction require separate checks.

Prediction: gains vary across windows

56-day block WIS · lower is better

	dev 1	dev 2	final
M0	0.1338	0.2046	0.1957
M1	0.1320	0.1893	0.3690
M2	**0.1265**	**0.1577**	0.3394

Paired seven-day block intervals, M2 minus M0:
dev 1: $[-0.0153, 0.0011]$; includes zero.
dev 2: $[-0.0675, -0.0231]$; excludes zero.

The pre-specified development-fold rule selects M2.
City-equal scores; one-step results in backup.

Final-window calibration

	M0	M2
Block WIS	0.1957	0.3394
90% coverage	75.5%	60.5%

Wider M2 intervals and lower coverage indicate a mismatch between predictive distributions and observations.

Final window previously inspected. No retuning after this evaluation.

Findings and their scope

01 Precipitation

About -20.21%

per $+10$ mm; negative in every city.

Stable across two full-period specifications, two contrast definitions and three dev-1 model fits.

02 Temperature

Positive full-period association

under both specifications.

Peak location changes with the specification. A single optimal temperature is not identified.

03 Time structure

Development gains have limits

Final M2 coverage: 60.5%.

Lower innovation correlation and better development scores; final block worse than M0. AR(1)'s separate contribution is unverified.

Observational associations; both prediction tasks condition on realised weather. Causal effects, climate adaptation and operational forecasting are outside the study scope.

Thank you · Questions

Backup 1: priors and sampling

Priors on the $\log(1+U)$ scale

Population intercept	$\mathcal{N}(7, 2)$
City intercept scale	HN(2)
UTCI, per 10°C	$\mathcal{N}(0, 0.5)$
Quadratic	$\mathcal{N}(0, 0.3)$
Rainfall	$\mathcal{N}(0, 0.3)$
Weekend	$\mathcal{N}(0, 0.4)$
Weekday contrasts	$\mathcal{N}(0, 0.4)$
Annual harmonic	$\mathcal{N}(0, 0.8)$
Trend, per 100 days	$\mathcal{N}(0, 0.3)$, HN(0.3)
AR persistence z	$\mathcal{N}(0, 0.5)$, HN(0.5)
Log noise scale	$\mathcal{N}(-0.5, 0.8)$, HN(0.5)

HN denotes a half-normal distribution.

Varying coefficients share the population-normal and half-normal group scale. Noise uses 0.8 for the population and 0.5 for the group.

Trend uses the fitted scale 0.3. Configuration field `beta_trend_city_sd`$= 0.2$ is unused.

M2 dev 1 full prior predictive check, 1,000 draws: 3.50% below zero; intercept-only check: 1.15%. The upper tail is wide. Source: `followup_checks.json` in `logs/review/`.

Acceptance: $\widehat{R} \leq 1.01$, bulk and tail ESS ≥ 400, no divergences, MCSE below 5% of the posterior standard deviation.

Backup 2: associations and prediction

Contrast (%)	M0	M2
Rain $+10$ mm	-19.98	-20.21
95% interval	$[-21.07, -18.89]$	$[-21.37, -18.99]$
Rain median–90th	-16.40	-16.99
95% interval	$[-17.28, -15.53]$	$[-18.02, -15.95]$
UTCI $+10°$C	5.22	6.16
95% interval	$[3.67, 6.76]$	$[4.12, 8.32]$
UTCI 25th–75th	19.89	14.04
95% interval	$[17.44, 22.47]$	$[11.11, 17.27]$
Interior peak	15/25	5/25
Outside range	1/25	1/25

City-equal percentages, 25 systems; 95% intervals. Peak: $P > 0.95$; outside range: UTCI $+10°$C.

Method (MAE)	Block	One-step
M0	0.3207	0.3207
Seasonal naive	0.4252	0.3421
City AR-OLS	0.5192	0.3197
City OLS	0.5208	0.5208
M2	0.5547	0.3601
City constant	0.5551	0.5551

Final city-equal MAE; lower is better. Naive: 7-day lag; OLS: parameter uncertainty. Baselines compare points only.

Final one-step WIS: M0 0.1957, M2 0.2085; 90% coverage: 75.7% and 72.0%, respectively.

Dev 1 temperature, 23 cities: M0 11.35% ($[9.72, 12.99]$), M1 10.07%, M2 7.49% ($[5.47, 9.65]$). This window and roster differ from the full-period fits.

Backup 3: limitations and reproduction

Limitations

- Student-t permits negative log usage. Expected usage does not exist for finite ν; report medians and quantiles.
- M0 leaves lag-one residual autocorrelation near 0.317.
- Sampling, model fit and prediction require separate checks.
- Both tasks condition on realised weather.
- The final block was inspected before this analysis.

Files

- Data: Mendeley Data, Version 1, CC BY 4.0. Cleaned CSV in `data/`; SHA-256 checked before fitting.
- Bean, Pojani and Corcoran (2021), *Journal of Transport Geography*, 95:103155.
- PyMC 6.3.2, nutpie 0.16.11; NUTS, four chains.
- Commands: `src/run.py`. Tests in `tests/`; verification: `src/verify_deliverables.py`.